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Measuring Agreement in Participatory Mapping

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Abstract

This article offers a way to measure agreement in participatory mapping. We asked subject matter experts (SMEs) to draw where Sudanese ethnic groups were located on a map. We then used an eigenanalysis approach to determine whether SMEs agreed on the location of ethnic groups. We used minimum residual factor analysis to assess the extent of interinformant agreement among SMEs' judgments of four ethnic groups' spatial locations. Although we found little evidence of the lumper/splitter problem, some SMEs drew larger polygons than others, and we provide directions on how to reduce this potential bias. Finally, the results of the geospatial agreement analysis were validated through a comparison of other sources of geospatial data for three of the four ethnic groups.

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Participatory mapping, which identifies the distribution of people, animals, and resources (Chambers 1994), is often collected in focus groups where some participants' opinions can dominate. This article uses an eigenanalysis approach to assess agreement of spatially explicit responses among respondents. This project was developed because conflict research requires data on ethnic group locations, and a great obstacle in conflict research is finding data at the appropriate spatial resolution (Buhaug and Lujala 2005). Since World War II, the number of civil wars has increased (Harbom and Wallensteen 2007), and conflict research has evolved from pairwise country comparisons (Bremer 1992) to intrastate analyses (Buhaug and Lujala 2005). Ethnic group distributions correlate with nonstate and intrastate conflict (Sambanis 2001; Van Holt et al. 2012), yet we lack finer resolution data for such studies.

Chronic ethnic conflict has plagued Sudan and South Sudan since independence in 1955 and has continued after their separation on July 9, 2011 (Lobban et al. 2002; Themner and Wallensteen 2012). Ethnic conflict persists over Abyei, an area along the Sudan and South Sudan border, resulting in the emigration and immigration of many ethnic groups from traditional locations (Ayers 2010; Themner and Wallensteen 2012).

Maps of Sudanese ethnic groups show extensive ethnic diversity, yet they lack validity assessments. Izady (2012) created one of the few spatially explicit ethnic group maps of Sudan and South Sudan by exhausting historical and anthropological information dating back to 1954. Murdoch (1959) produced a widely recognized ethnic group map of Africa. A recent map available online (Muturzikiña 2007) combined historical atlases, Ethnologue (Lewis et al. 2013), and the *World Atlas of Language Structures* (WALS; Forkel 2011).

Ethnic group locations could be collected and verified on site. However, countries such as Sudan and South Sudan are so large in area (2,505,810 square kilometer—the size of Alaska, Texas, and Maine combined) and population (60,602,000 projected for 2025; Lobban et al. 2002) that on-site mapping would be an enormous undertaking. An alternative we suggest is participatory mapping where ethnic group locations drawn by multiple subject matter experts (SMEs) are combined, validated, and compared. To explore the nature of agreement among informants, we use an eigenanalysis approach, specifically minimum residual factor analysis, to determine the extent to which there is shared understanding among respondents' determinations of ethnic locations.

Background

Assessing Shared Agreement of Ethnic Group Boundaries

We want to determine whether there is a general shared understanding of perceived ethnic group locations among a set of informants, here SMEs. Our use of an eigenanalysis approach is conceptually similar to that of the cultural consensus model (CCM; Romney et al. 1986) for assessing the extent to which there is a shared cultural understanding among a sample of informants or the use of minimum residual factor analysis or the use of a singular value decomposition for determining core–periphery structure in valued networks (Boyd et al. 2010). However, we advocate this as a more exploratory or heuristic procedure for determining agreement in informants' geospatial perceptions and is in no way based on any formal axiomatic foundation. Thus, the article provides an example an extension of eigenanalysis to geospatial similarity data.

Method

Ethnic Group Sample

The names of 25 ethnic groups at approximately the second level were provided by Sudanese experts, Richard and Carolyn Lobban. Ethnic groups have three levels. For example, the Dinka (first level) contains smaller, regionally localized ethnic branches, such as the Ngok, Malwal, and Bor (second level), which themselves contain family groups (third level) based on genetically and socially unique lineages. The ethnic branches had enough groups for the consensus analysis method, yet the groups are large enough to be known at a regional level. The final list includes Azande, Bari, Beja, Beni Amer, Bor, Danagla, East Jikany, Hadendowa, Humr, Ja'aliyin, Kakwa, Lou, Madi, Malwal, Masalit, Messirya, Murle, Ngok, Rizaygat, Shaygiya, Shilluk, Taisha, Talodi, West Jikany, and Zaghawa. After the first four interviews, we added the Fur, who were inadvertently excluded.

Selecting Respondents

We interviewed 17 experts attending the Sudanese Studies Conference in Tempe, Arizona (May 18–20, 2012). To identify the experts, conference organizers identified key informants (J. C. Johnson 1990). We also distributed flyers and invited participants to our round table discussion of, participatory understanding of Sudanese ethnic groups.

Response and Reference Maps

Response and reference maps of Sudan and South Sudan were created. A response map recorded hand-drawn perceptions of ethnic group location boundaries. The response map included state borders, key cities, roads, rivers, and mountain ranges (defined as areas above 800 m, from a digital elevation model; Food and Agricultural Organization (FAO) 2004; Lehner et al. 2006; United Nation Development Program 2010). Each response map contained an ethnic group label and a confidence scale (1–5) for the drawn polygons where five represents very confident and one represents not very confident. Sudanese experts Loban and Fluehr-Lobban evaluated a preliminary response map to identify key features to include, since in scaled mapping, respondents orient themselves within the map (Rambaldi et al. 2006). The reference map provides an additional land cover and political boundary information (e.g., state boundaries); instead of the digital elevation model, a Landsat satellite image that represented land cover was used (U.S. National Park Service 2009). After a few interviews, we provided a state name map.

Interviews and Data Processing

Respondents were independently interviewed and asked to draw the ethnic group locations. Each ethnic group had a separate response map and each group was recorded separately. All responses were voluntary; some respondents did not respond about certain ethnic groups because they did not recognize the name.

The hand-drawn response maps were scanned in at 200 dots per inch and georeferenced using four control points. The average root mean square error for all 398 maps was 6.9 ± 0.75 km and ranged from .02 km to 10.96 km. The response polygons were transformed to a digital vector file using ArcScan (stet ArcScan), which were then converted into a raster layer (pixel size of 1.28 sq km). The rasterized polygon that represented the ethnic group boundary was assigned a value of 1 and the remaining cells (the rest of Sudan and South Sudan) were assigned 0s. The raster layer was transformed to Africa Albers equal area conic projection with the World Geodetic System (WGS) 1984 datum to reduce area distortion.

Analysis

Geospatial Agreement Areas

We integrated all 17 of the respondents' raster layers for each of the 26 ethnic groups into an ethnic agreement raster layer (EARL). EARLs reflect

the number of respondents that agree on ethnic group presence at a given pixel. Each raster layer contains many 1 or 0 valued cells. One indicates the respondent perceived that an ethnic group is present at that pixel, while zero indicates absence. The cell statistics function in ArcGIS (10.0) summed all the respondents' raster layers together for each ethnic group. The final pixel values represent how many respondents identified that pixel as containing an ethnic group. For example, if three experts agree that the Dinka were present in pixel (x, y), its value would be 3. We included ethnic groups that had at least 10 respondents agreeing on at least some pixels for the agreement analysis, which included the Ngok, Danagla, Ja'aliyin, and Shilluk, because this provided the widest range of agreement values for analysis (EARLs ranged from 0 to 13).

Individual responses were transformed into an aggregated asymmetrical proximity matrix made up of pairwise percentages of *i*s overlap with *j* and *j*s overlap with *i*. This agreement matrix contained the averaged percentage of area overlap between each pair of respondents. To calculate the percentage area overlap between a pair of respondents, the area of intersection between the pair was divided by the total area of each corresponding response polygon (Figure 1).

This results in a different percentage for each respondent and an asymmetrical matrix (i.e., *i*s overlap with *j* and *j*s overlap with *i*) because response polygons vary in size. For example, the first and third respondents had an intersection area of 27,023 sq km and individual areas of 31,564 sq km (respondent 1) and 40,391 sq km (respondent 3; Figure 1). The first respondent's polygon overlaps with 67% of the third respondent's polygon, while the third respondent's polygon overlaps with 86% of the first respondent's polygon (Figure 1). The response-polygon area affected the percentage; the first respondent's polygon had a smaller area, and thus had a higher percentage of overlap. Since the minimum residual factor analysis requires a symmetrical matrix, the percentage of overlap between each pair of respondents was symmetrized by averaging the overlap (77%). We refer to this method for determining interinformant agreement as the average method.

There are a variety of other possible methods for determining spatial agreement in this asymmetric case. An alternative method would be to calculate the total area of overlap between pairs of respondents relative to the total amount of area across both informants. For the geospatial agreement between *i* and *j*, this would be simply $S_{ij} = 2(O_{ij})/(a_i + a_j)$, where S_{ij} is the geospatial agreement between two informants, O_{ij} is the area of overlap between them, and a_i and a_j are the total areas for each respondent,

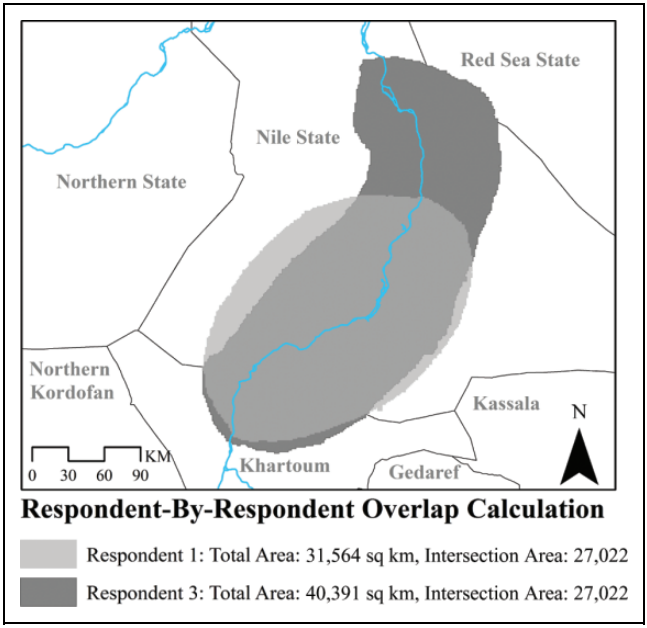

Figure 1. Example of the respondent-by-respondent overlap calculation used for the analysis.

referred to as the area method for determining interinformant agreement. In the eigenanalysis, we apply both methods in testing for geospatial agreement. It is important to point out, however, that both methods resulted in similar results. Quadratic assignment procedure correlations between the aggregate matrices for each group were .912 ($p = .002$) for the Shiiluk, .905 ($p = .0002$) for the Danagla, .866 ($p = .0002$) for the Ja'aliyin, and .875 ($p = .0002$) for the Njok (Hubert and Schultz 1976).

Geospatially Applying the CCM

To determine whether the respondents' overlapping responses indicate shared knowledge, we used an exploratory eigenanalysis approach. Agreement reflects the extent that each pair agrees with one another that a given ethnic group is present at the same area. A minimum residuals factor analysis was run on the respondent-by-respondent agreement matrix using both the average and area methods for determining interinformant agreement. We note that this is a very informal approach where we are simply applying

an exploratory descriptive procedure, minimum residual factor analysis, to a descriptive set of data, the interinformant agreement in terms of SMEs' judgments of ethnic groups' spatial locations.

We further note that there is currently no formal axioms for this approach although we hope to address this in future work. Also whereas eigenanalysis techniques, such as minimum residual factor analysis, are usually applied to correlations and covariance matrices, there are numerous examples of their use with similarity data. For example, valued adjacency data in social networks (Borgatti and Everett 1999; Boyd et al. 2010; J. C. Johnson et al. 2003), Euclidean similarity data in meteorology (Elmore and Richman 2001), and genetic similarity data for population identification (Lawson and Falush 2012) are all examples of eigenanalysis techniques applied to similarity matrices. There are several cautions in using eigenanalysis with similarity data concerning such things as the distribution of similarity values (e.g., skewness) and whether the matrix is positive definitive, but these issues are beyond the scope of this article.

In applying minimum residual factor analysis to interinformant agreement data, we judge sufficient agreement among the SMEs by the extent to which the analysis yields a single-factor solution. In other analogous approaches, like the CCM, this is generally determined by the magnitude of the first to second eigenvalues and by the sign and magnitude of the first factor scores (Weller 2007). The rule of thumb among many applications of the CCM in the literature is that there is evidence for consensus or agreement if the ratio between the first and second eigenvalues is greater than 3:1 (Weller 2007). However, in more recent personal communications with some of the original authors of the CCM, Susan Weller and William Batchelder, this rule of thumb should be supplemented. Even when very high ratios of the first to second eigenvalues are present, information and knowledge found in the second factor may still remain (see Boster and Johnson 1989). As a result, we employed eigenvalue scree plots to assess how well the data reflect a single-factor solution by visualizing the magnitude of the first eigenvalue in relation to subsequent eigenvalues.

Since we are doing a factor analysis on the SMEs, the first factor scores in the minimum residual factor analysis represents the extent to which each respondent geospatially agreed with the aggregate. The scores can range from -1 to 1, but to reflect agreement, they should all be positive and relatively large (greater than .50; Weller 2007).

Given that there was no constraint on the size of a respondent's polygon, we wanted to make sure that agreement and competency was not biased by polygon size. This problem in the effect of polygon size on the results is

similar to the lumper/splitter problem in pile sort analysis (Bernard 2013). To test if polygon size influences the first factor scores, a bivariate Pearson's two-tailed correlation analysis was run comparing respondents' individual polygon size with their competency scores for each ethnic group. The analysis found no significant correlations between size and competency across all four groups, and we can feel more confident that this is not biasing the results of the agreement analysis.

Results

Is There a Consensus among the SMEs about the Locations of Ethnic Groups?

The responses showed consistent consensus for the locations of the Ja'aliyin and the Shilluk ethnic groups across the two methods for determining agreement. The responses for the Ngok and Danagla ethnic groups still seem to point to some degree of a shared understanding, though there was some variation in the results obtained from the two methods (Figure 2).

The Ja'aliyin had a ratio between the first and second eigenvalues in the minimum residual factor analysis of 4.89 and 4.03 for the average and area method, respectively (Table 1).

For the Shilluk, the ratio was 3.18 for the average method (3.48 for area method), both indicating agreement among respondents. Results for the remaining two analyses had slightly different results with respect to the two methods for determining agreement. The Dangala ratio was 2.89 and 3.38 and the Ngok yielded a ratio of 2.48 and 3.15 for the average and area method, respectively.

The average across all respondents' first factor scores for each ethnic group ranged from .71 for the Ja'aliyin responses to .52 for the Danagla respondents (Table 1), indicating agreement within the data. The average first factor scores for individual respondents for the average method ranged from .35 (respondent 26) to .77 (respondent 1), with no negative values. Thirteen of the 17 respondents (Online Appendix) met or exceeded the average factor score threshold of .50, to indicate reasonable agreement.

The scree plot of the first three eigenvalues for the four ethnic groups for the average method (not shown) showed stronger signs of agreement given both the eigenvalue ratio and the flattening of the line in the scree plot between the second and third eigenvalues. Of the four, the Ngok had the most amount of residual knowledge indicated by the steeper decline of the line between the second and third eigenvalues. The informants had the most

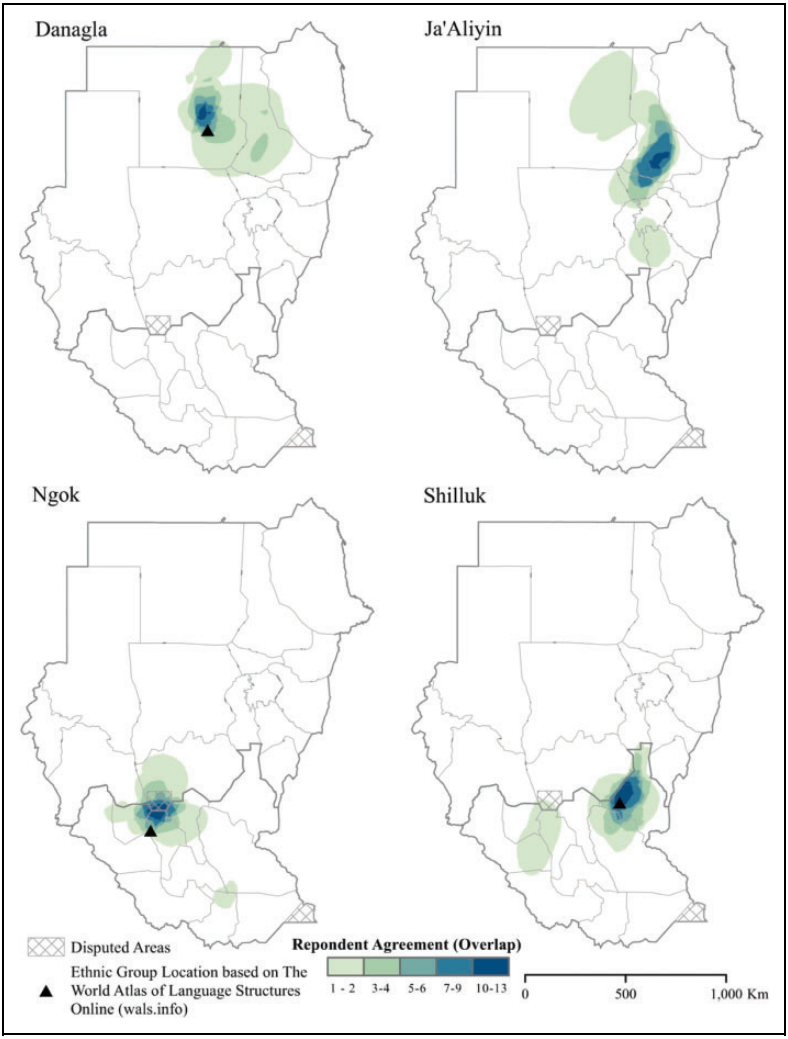

Figure 2. Ethnic agreement raster layer maps for the four ethnic groups analyzed (reflecting the number of overlapping respondents). Triangles represent latitude and longitude locations derived from the *World Atlas of Language Structures*.

(13) overlapping response polygons for the Ngok; however, the Ngok had the lowest eigenvalue ratio. In sum, there appears to be agreement across all four ethnic groups although some are greater than others.

Table 1. Results from a Minimum Residual Factor Analysis Using Both Methods for Determining Informant Overlap—Generating Average and Percentage Area Relative to Total (in Parentheses).

Ethnic Group	Ja'aliyin	Shilluk	Njok	Danagla
No. of negative first factor scores	0	0	0	0
Largest eigenvalue	7.38 (5.96)	6.64 (5.94)	5.86 (4.89)	5.37 (4.29)
Second largest eigenvalue	1.51 (1.48)	2.08 (1.71)	2.36 (1.55)	1.86 (1.27)
Ratio of largest to next	4.89 (4.03)	3.18 (3.48)	2.48 (3.15)	2.89 (3.38)

Discussion

Orientation and Livelihood

The informants agreed more about the locations of the Ja'aliyin and the Shilluk (Table 1). This could be because both groups are located near a clearly mapped large city and river. The Ja'aliyin were perceived to be just north of Khartoum, the capital of Sudan, and along the Nile River. The Shilluk were perceived to be located in and north of Malakal, a large city in the north of South Sudan along the White Nile River.

The informants had slightly more variation in agreement for the Danagla and Ngok (Table 1). The Danagla were perceived to be located along the Nile River, but there is no urban center to help orient respondents. The respondents placed this group along the Nile, north of Khartoum, and all but three placed them in the northern state (Ash Shimaliyya) near its capital of Dunqulah, which was not marked on the response map. The Ngok were perceived to be located in the southern half of a disputed area along the Sudan/South Sudan border in the state of southern Kordofan and extending into South Sudan states of Warab, Unity, and Northern Bahar. Unrest in the area could have lowered the level of agreement. However, since the Ngok had the highest number of overlaps, there is a reasonable level of agreement about their location.

Livelihoods could help explain some of the variation in agreement. The Ja'aliyin and the Shilluk are mostly sedentary, while the Shilluk and the Ngok are less so, which might contribute to variations in agreement. In the 19th century, the Ja'aliyin were merchants and traders, but today they mostly herd camels and cattle locally (Lobban et al. 2002). The Shilluk are farmers who fish and own livestock and dairy cows (Lobban et al. 2002). On the other hand, the Danagla are migrant merchants who trade all over Sudan and South Sudan (Lobban et al. 2002). The Ngok have been settled for over two centuries in Abyei (D. H. Johnson 2008);

however, this area is disputed between Sudan and South Sudan (Themner and Wallensteen 2012).

Triangulation with Other Data Sources

To assess the validity and accuracy of the ethnic group locations, we compare our EARLs to other sources. We use spatial locations from WALS for language groups that can be directly compared to the ethnic groups in the study. Latitude and longitude coordinates were provided for the Ngok, Danagla, and Shilluk since these ethnic groups are associated with a language group of the same name. The Ja'aliyin are Arabic speakers, so a comparison of this kind is not reasonable, given the predominance of Arabic in the northeast region of Sudan. The latitude and longitude locations from WALS (up triangles) are at or near the center of the EARLs in each case (Figure 2). This triangulation of the two separate data sets lends support for the validity of the geospatial approach we advocate here.

Recommendations for Future Research

The open-ended nature of our drawing instructions could have been a factor in the variability in the size of respondents' drawn polygons, possibly influencing levels of the agreement. Using more specific drawing instructions (e.g., "designate the center of an ethnic group location and then draw a polygon for the full extent of that same ethnic group," or "draw the location where you believe 80% of the population resides") could help standardize the geospatial responses and possibly increase agreement. It may also be interesting to quantify and analyze the area where respondents fail to overlap with each other. Orienting the respondents proved problematic within states that did not have major features such as large rivers or main roads. Even though it may clutter the response map, the state capital locations could help orient the respondent within states. Finally, soliciting interviews from experts with more diverse backgrounds and experiences could help capture the full extent of responses. With the presumed increase in agreement, a scree plot of additional unique polygons by area can help determine when adding additional people no longer improves the map.

In future research, alternative calculations could be compared in vector and raster environments. Here we compared two methods for calculating overlap (average and percent area relative to total), and there could be other

methods in the geographic information system (GIS) or remote sensing software to run similar calculations. One alternative could involve raster data set calculators to generate an informant-by-informant agreement matrix for each pixel, which could be used to create an aggregate proximity matrix (using positive matches). A factor analysis could be run on this aggregate proximity matrix, although one potential limiting factor could be the sheer number of pixels involved (i.e., millions). This approach may also help solve some of issues of nonindependence in the geospatial perception data that may need to be considered and is analogous to issues of nonindependence in determining interinformant agreement in cognitive social structure data in social network analysis. This is something that will be considered in more detail in future work. We also hope in subsequent work to make these analyses spatially explicit in a GIS environment. This can bring together eigenanalysis, which has a long history, with geospatial analysis, to formalize assessments of agreement in geospatial data and to understand the etic characteristics that explain variation in agreement and cultural geospatial knowledge.

Conclusion

This research focused on how to synthesize and gain meaning from multiple geospatially explicit responses. Being able to aggregate and add meaning to geospatially explicit perceptions of multiple SMEs enables researchers to produce spatial data that are current and incorporate multiple perspectives at a relatively rapid pace. Although there is still much to understand about this approach, maps created using this method have the potential to be powerful tools for eliciting geospatial information and assessing its reliability.

Declaration of Conflicting Interests

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Supplemental Material

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